SECURITISATION AI AGENT

Financial Risk Assessment Dashboard
Using Advanced DAX Analytics in Power BI

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EXECUTIVE SUMMARY

PROJECT OVERVIEW

This report presents the complete methodology, analysis, and deliverables of a 20-day financial data analytics internship focused on securitisation risk assessment using Microsoft Power BI and advanced DAX (Data Analysis Expressions).

The project involved building an enterprise-grade risk assessment platform for an auto loan securitisation pool consisting of 500 loans with a total outstanding balance of ₹317.79 million.

Working with four simulated datasets representing data typically extracted from SAP/ERP systems, the project demonstrates how a data analyst transforms raw loan-level data into actionable risk intelligence for investors, rating agencies, and regulators.

POOL OVERVIEW (Key Numbers)

Total Outstanding Balance: ₹317.79 Million
Original Pool Balance: ₹546 Million
Active Loans: 500
Pool Factor: 0.58 (58% remains)
Weighted Average Coupon: 10.95%

RISK METRICS

30+ DPD Rate: 16%
NPA Rate (90+ DPD): 12%
Total NPA Loans: 23
30+ DPD Loan Count: 167
60+ DPD Loan Count: 70
Collection Efficiency: 77% (current loans)

IFRS 9 ECL SUMMARY

Stage 1 (Healthy): 422 loans = 84.4% of pool
Stage 2 (Stressed): 45 loans = 9.0% of pool
Stage 3 (Defaulted): 33 loans = 6.6% of pool
Stage 1 ECL Provision: ₹2.79 Million

Stage 2 ECL Provision: ₹1.65 Million
Stage 3 ECL Provision: ₹7.22 Million
Total ECL Provision: ₹11.66 Million
ECL Coverage Ratio: 4% of pool balance

KEY FINDING

The portfolio shows elevated delinquency levels with 30+ DPD at 16%, significantly exceeding the standard trigger threshold of 5%.

The DPD trigger test is currently FAILING. Stage 3 (NPA) loans represent 6.6% of the pool requiring ₹7.22M in provision alone.

Immediate action recommended:
→ Increase recovery efforts on 60+ DPD loans
→ Review underwriting standards for new loans
→ Monitor roll rates from 30 DPD to 60 DPD
→ Prepare investor communication on DPD status

DASHBOARDS DELIVERED
5 interactive Power BI dashboards were built:

Dashboard 1: Executive Summary
→ Complete portfolio health overview
→ 8 KPI cards, DPD distribution, state analysis

Dashboard 2: DPD Analysis
→ Delinquency monitoring and roll rate matrix
→ Monthly trend analysis with moving average

Dashboard 3: Vintage Analysis
→ 15 vintages tracked with CNL curves
→ 12-month and 24-month comparison charts

Dashboard 4: IFRS 9 ECL
→ Complete Stage 1, 2, 3 classification
→ ECL computation: PD × LGD × EAD

Dashboard 5: Cash Flow Waterfall
→ Monthly cash flow allocation
→ OC/IC/DPD/Loss trigger compliance testing

PART A: TRAINING MATERIAL AND LEARNING RESOURCES

A1. SECURITISATION FUNDAMENTALS

What is Securitisation?

Securitisation is the financial process of converting illiquid loan assets into tradeable securities. A bank bundles hundreds of individual loans together, transfers them to a Special Purpose Vehicle (SPV), which then issues securities to investors backed by the cash flows from those loans.

The process solves a fundamental banking problem: banks have limited capital but unlimited demand for loans. By securitising existing loans, banks convert future EMI income into today's cash, allowing them to give more loans.

How Securitisation Works - 7 Steps:

Step 1: Bank gives 500 auto loans to borrowers
(Each borrower pays EMI every month)

Step 2: Bank bundles all 500 loans together
(Total pool = ₹317.79 Million)

Step 3: Bank creates a Special Purpose Vehicle
(SPV is a separate legal company)

Step 4: Bank transfers all loans to the SPV
(SPV now legally owns the loans)

Step 5: SPV issues securities to investors
(Securities divided into risk tranches)

Step 6: Investors pay cash to buy securities
(Bank receives ₹317.79M in fresh cash)

Step 7: Borrowers pay EMI every month
(Flows through SPV to investors)

Key Participants:

Originator: Bank that gave the original loans
Example: HDFC Bank, Bajaj Finance

SPV: Special Purpose Vehicle
Separate legal entity holding loans
Protects investors if bank fails

Servicer: Collects EMIs from borrowers
Usually the same bank (originator)

Trustee: Oversees the SPV and distribution
Example: IDBI Trusteeship

Rating Agency: Rates each security tranche
India: CRISIL, ICRA
Global: S&P, Moody's, Fitch

Investors: Buy the securities
LIC, Mutual Funds, Pension Funds

Data Analyst: Monitors portfolio and reports
This is my role in this project

Tranche Structure:

Senior (AAA): 70% of pool
Safest, lowest return (8%)
Paid FIRST from collections
Loses money LAST

Mezzanine (AA): 20% of pool
Medium risk, medium return (11%)
Paid SECOND

Equity: 10% of pool
Riskiest, highest return (20%+)
Paid LAST
Absorbs losses FIRST
Usually kept by the bank

Credit Enhancement:
Losses hit Equity first, then Mezzanine,
then Senior LAST. Senior investors are protected
by 30% credit enhancement (equity + mezzanine).

Types of Risk in Securitisation:

Credit Risk: Borrower does not pay EMI
Measured by DPD and NPA rates

Prepayment Risk: Borrower pays early
Investor receives money sooner
than planned (reinvestment risk)

Concentration: Too many loans in one state
or one vehicle type
One event can crash entire pool

Interest Rate: Market rates change unexpectedly
Affects floating rate pools

Structural Risk: OC test or IC test fails
Cash flow gets redirected

Key Terms Explained:

DPD (Days Past Due):
How many days late is the EMI payment?
0 DPD = Paid on time (GOOD)
30 DPD = 30 days late (WARNING)
90+ DPD = 90+ days late = NPA (DEFAULT)

NPA (Non Performing Asset):
RBI rule: Any loan 90+ days late = NPA
Bank must set aside heavy provisions for NPAs
Cannot recognize interest income on NPAs

CIBIL Score:
Credit score of borrower (300 to 900)
800+ = Excellent (very low risk)
700-750 = Good (acceptable)
Below 650 = Poor (high risk)
LTV (Loan to Value):
LTV = Loan Amount ÷ Vehicle/Asset Value

LTV 75% = Bank lent 75% of vehicle value
High LTV = Less collateral protection

WAC (Weighted Average Coupon):
Average interest rate of entire pool
weighted by loan balance.
Your pool WAC = 10.95%

ECL (Expected Credit Loss):
Money bank must set aside for expected losses.
ECL = PD × LGD × EAD
Your pool ECL = ₹11.66 Million (4%)

OC (Overcollateralisation):
Pool Balance ÷ Senior Tranche Balance
Must stay above 120% (trigger level)

Waterfall:
The order in which cash is distributed.
Senior investors paid first, equity last.

RBI Regulatory Framework:

MRR (Minimum Retention Requirement):
Bank must keep at least 10% of each tranche.
Cannot sell everything and walk away.
Ensures bank has "skin in the game."

MHP (Minimum Holding Period):
Bank must hold loans for at least 12 months
before securitising them.
Prevents immediate securitisation of bad loans.

SMA (Special Mention Account):
RBI early warning system.
SMA-0: 1-30 days late
SMA-1: 31-60 days late
SMA-2: 61-90 days late
NPA: 90+ days late

NACH Mandate:
National Automated Clearing House.
Auto-debit system for EMI collection.
Higher NACH coverage = more reliable collections.

A2. DATA REQUIREMENTS

Overview of Datasets Used:

This project used four Excel datasets representing loan-level and pool-level data typically extracted from SAP/ERP systems at a bank.

Dataset 1: Auto Loan Securitisation
Purpose: Core loan-level data for entire pool
Records: 500 loans
Key Columns:
→ LoanID: Unique identifier for each loan
→ BorrowerID: Unique borrower identifier
→ OriginalBalance: Loan amount at origination
→ CurrentBalance: Outstanding balance today
→ InterestRate: Annual interest rate on loan
→ DPD_Days: How many days past due
→ CIBIL_Score: Borrower credit score
→ LTV_Ratio: Loan to value ratio
→ State: Geographic location of borrower
→ VehicleType: Car, SUV, Commercial etc
→ OriginationDate: When loan was given
→ EmploymentType: Salaried or Self-Employed
Used in: Dashboards 1, 2, 4, 5

Dataset 2: DPD Snapshot History
Purpose: Monthly delinquency records
Key Columns:
→ SnapshotDate: Month end date of record
→ LoanID: Links to AutoLoans table
→ DPD_Days: Days past due that month
→ DPD_Bucket: Bucket category (Current, 1-29 etc)
→ DPD_Bucket_Prior: Bucket in previous month
→ CurrentBalance: Balance at that snapshot
Used in: Dashboard 2 (Roll Rate and DPD trend)

Dataset 3: Static Pool Vintage Data
Purpose: Track loan cohorts from origination
Key Columns:
→ VintageID: Quarter of origination (2021-Q1 etc)
→ MonthsOnBook: Months since origination (0-60)
→ OriginalPoolBalance: Starting balance of vintage
→ RemainingPoolBalance: Balance still outstanding

→ CumulativeNetLoss: Total losses since start
→ CumulativeNetLossRate: Loss as % of original
Used in: Dashboard 3 (Vintage Analysis)

Dataset 4: Dynamic Loss Monthly
Purpose: Monthly pool summary performance
Key Columns:
→ ReportingDate: End of month date
→ BOP_Balance: Balance at start of month
→ EOP_Balance: Balance at end of month
→ CollectionsTotal: Total EMIs collected
→ GrossLoss_ThisMonth: Losses this month
→ Recoveries_ThisMonth: Recoveries this month
→ NetLoss_ThisMonth: Gross loss minus recoveries
→ Prepayments_ThisMonth: Early payoffs
→ ScheduledAmort: Normal principal reduction
Used in: Dashboard 5 (Waterfall)

Data Model Architecture:

The star schema connects all tables through
a central DimDate calendar table.

DimDate (Central Hub - Dimension Table):
Contains every calendar date from 2019 to 2027.
Year, Month, Quarter, MonthYear columns.
Marked as Date Table for time intelligence.

5 Relationships Built:
1. DimDate → AutoLoans (via OriginationDate)
2. DimDate → DPDSnapshot (via SnapshotDate)
3. DimDate → DynamicLoss (via ReportingDate)
4. DimDate → StaticPool (via VintageStartDate)
5. AutoLoans → DPDSnapshot (via LoanID)

All relationships are One-to-Many (1 to *)
DimDate is always on the "1" side.

A3. DAX CONCEPTS LEARNED

DAX (Data Analysis Expressions) is the formula
language used in Microsoft Power BI, Analysis
Services, and Power Pivot in Excel.

Core Concept: Filter Context
Every DAX measure evaluates within a filter context.
When a slicer filters to Maharashtra, all measures automatically show only Maharashtra data.
This is the most important concept in DAX.

Key DAX Patterns Mastered:

Pattern 1: SUM and COUNT
Basic aggregation functions.
Total Balance = SUM(AutoLoans[CurrentBalance])
Total Loans = COUNTROWS(AutoLoans)
These are the foundation of all measures.

Pattern 2: CALCULATE
Most important DAX function.
Evaluates an expression after modifying filters.

Example:
```
NPA Balance =
CALCULATE(
    SUM(AutoLoans[CurrentBalance]),
    AutoLoans[DPD_Days] >= 90
)
```
This calculates balance ONLY for 90+ DPD loans.

Pattern 3: DIVIDE
Safe division that handles zero denominators.
Always used instead of the / operator.

```
NPA Rate =
DIVIDE(
    CALCULATE(SUM(Balance), DPD>=90),
    SUM(Balance)
)
```
Returns blank instead of error if denominator = 0.

Pattern 4: SUMX (Iterator Function)
Calculates an expression for each row then sums.
Used for weighted average calculations.

```
WAC =
DIVIDE(
    SUMX(AutoLoans,
        AutoLoans[CurrentBalance] * AutoLoans[InterestRate]),
    SUM(AutoLoans[CurrentBalance])
)
```
Multiplies balance × rate for EACH loan, then sums.
Divides by total balance = weighted average rate.

Pattern 5: VAR (Variables)
Makes complex formulas readable and reusable.
Calculates expression once, uses multiple times.

```
NPA Rate with VAR =
VAR NPA_Balance =
    CALCULATE(SUM(Balance), DPD >= 90)
VAR Total_Balance =
    SUM(Balance)
RETURN
    DIVIDE(NPA_Balance, Total_Balance)
```

Pattern 6: SWITCH
Multiple condition branching.
Used for IFRS 9 stage classification.

```
IFRS9_Stage =
SWITCH(
    TRUE(),
    AutoLoans[DPD_Days] >= 90, "Stage 3",
    AutoLoans[DPD_Days] >= 30, "Stage 2",
    "Stage 1"
)
```

Pattern 7: Time Intelligence
Functions that work with dates for trend analysis.

```
Prior Month DPD =
CALCULATE(
    [30Plus DPD Rate],
    PREVIOUSMONTH(DimDate[Date])
)
```

```
3 Month Rolling Average =
AVERAGEX(
    DATESINPERIOD(DimDate[Date], MAX(DimDate[Date]), -3, MONTH),
    CALCULATE([30Plus DPD Rate])
)
```

Pattern 8: What-If Parameters
Interactive sliders connected to measures.
Used in stress testing scenarios.
Slider value multiplies base measures.

```
Stressed NPA =
[NPA Rate] *
'Default Rate Multiplier'[Default Rate Multiplier Value
```

PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

B1. SIMULATION CONCEPT AND PROBLEM STATEMENT

Platform Name: Securitisation Risk Arena

The Problem Being Solved:

India's securitisation market processes over
₹1.8 lakh crore annually. The data analysts
who monitor these portfolios are typically
trained by working on live portfolio data.

This creates a serious risk:
→ One wrong ECL formula = bank under-provisioned
by crores, causing regulatory penalties
→ One missed trigger breach = investor legal
dispute and rating downgrade
→ One incorrect waterfall calculation = wrong
investor payment creating legal liability

There is no standardised safe practice environment
for securitisation analytics training in India.

The Solution:

Securitisation Risk Arena is a browser-based
practice platform where analysts train on
simulated duplicate data in a game environment.

Mistakes cost game points, not real money.
Skills are built before touching real portfolios.

Analogy: Exactly like a flight simulator.
Pilots crash simulators hundreds of times safely.
Then they fly real planes with full confidence.

Securitisation Risk Arena = Flight simulator
for bank securitisation data analysts.

Platform Concept:

The player assumes the role of Lead Data Analyst
at Meridian Structured Finance Bank.

Portfolio managed: ₹5,000 crore securitised
auto loan pool across 6 fictional structures.

Each game session = One simulated business day.
Morning: Receive portfolio health alerts.
Afternoon: Build dashboards and run analysis.
Evening: Present findings to Chief Risk Officer.

The data used is simulated/duplicate data
structured exactly like real bank data
but containing no real borrower information.

Connection to This Project:

The four Excel datasets used in this project
(Auto Loans, DPD Snapshot, Static Pool,
Dynamic Loss) are exactly the type of
simulated data the game platform would use.

Dashboard 5 (Waterfall) in this project
demonstrates the trigger monitoring concept
that forms Level 5 of the game.

B2. GAME LEVELS AND MECHANICS
Campaign Mode: 8 Progressive Levels

Level 1: Data Foundation
Difficulty: ★☆☆☆☆ (Beginner)
Story: First day at work. Manager needs
basic pool metrics by lunch. 45 minutes.
Task: Calculate pool balance, DPD counts,
basic pool statistics.
DAX Skills: SUM, COUNT, CALCULATE, DIVIDE
Time: 45 minutes
Points: 100

Level 2: The Time Machine
Difficulty: ★★☆☆☆ (Easy)
Story: Rating agency wants 12 months of
DPD trend data. Meeting in 2 hours.
Task: Build monthly trend, rolling averages,
period-over-period comparisons.
DAX Skills: PREVIOUSMONTH, DATESINPERIOD,
AVERAGEX, DATESYTD
Time: 60 minutes
Points: 150

Level 3: The Weighted World
Difficulty: ★★★☆☆ (Medium)
Story: S&P needs weighted averages and
concentration analysis for deal rating.
Task: WAC, WAM, WA LTV, WA CIBIL,
HHI concentration index, TOPN analysis.
DAX Skills: SUMX, AVERAGEX, TOPN, RANKX
Time: 90 minutes
Points: 200

Level 4: Virtual Table Master
Difficulty: ★★★☆☆ (Medium)
Story: CRISIL wants 6 stratification tables.
Senior analyst just quit. You are alone.
Task: Pool stratification by LTV band,
CIBIL band, state, vintage, employment type.
DAX Skills: SUMMARIZE, ADDCOLUMNS,
CALCULATETABLE, REMOVEFILTERS
Time: 120 minutes
Points: 250

Level 5: Waterfall Wizard
Difficulty: ★★★★☆ (Hard)
Story: Investor claims wrong payment.
Find waterfall error before legal team arrives.
Task: Complete cash flow waterfall,
OC test, IC test, excess spread.
DAX Skills: Sequential allocation logic,
nested CALCULATE patterns
Time: 120 minutes
Points: 250

Level 6: ECL Engineer
Difficulty: ★★★★★ (Hard)
Story: RBI examiner arrives tomorrow
for IFRS 9 audit. Must get ECL right tonight.
Task: Stage classification, PD/LGD/EAD
computation, crosscheck against Excel.
DAX Skills: SWITCH, SUMX, FILTER,
cross-validation techniques
Time: 150 minutes
Points: 300

Level 7: Rating Agency Reporter
Difficulty: ★★★★★ (Expert)
Story: Simultaneous review by S&P,
Moody's, and CRISIL. Data tapes due by 8 AM.
Task: Complete data tapes for all 3 agencies,
all required fields populated and validated.
DAX Skills: All patterns combined
Time: 180 minutes
Points: 350

Level 8: CRISIS MODE
Difficulty: ★★★★★ (Extreme)
Story: It is 2008. All 6 pools deteriorating.
CRO on phone. Board wants answers.
Investors calling. 60 MINUTES to save the bank.
Task: Identify all trigger breaches, stressed ECL,
crisis dashboard, CRO presentation.
DAX Skills: All skills under time pressure
Time: 60 minutes
Points: 500 (Bonus level)

Scoring System:

EARN POINTS:
Correct DAX formula: +100 pts
Waterfall allocation correct: +200 pts
ECL matches Excel crosscheck: +150 pts
All measures under 100ms: +150 pts
Early warning detected: +50 pts bonus
Dashboard quality: +100 pts

LOSE POINTS:
Hard-coded value in formula: -50 pts each
Wrong waterfall allocation: -25 pts per step
Missed trigger breach: -30 pts
ECL stage classification wrong: -20 pts per stage
Measure exceeds 100ms: -10 pts each

MAXIMUM SCORE: 1000 points per session

GRADE SCALE:
900-1000: Distinction (Hire immediately)
800-899: Merit (Strong candidate)
650-799: Pass (Needs more practice)
Below 650: Fail (Remedial training)

Achievement Badges (9 Total):
DAX Master: Write 50+ correct measures
Waterfall Wizard: Zero misallocation errors
ECL Engineer: 0.01% variance from Excel
Crisis Survivor: Level 8 with Distinction
Early Warning: Caught 3 signals early
Speed Demon: All measures under 50ms
CRISIL Star: Perfect rating agency data tape
Vintage Veteran: 5 vintage analyses complete
Chief Risk Officer: Maximum player rank reached

Player Progression (8 Ranks):
Rank 1: Data Analyst Trainee (0 XP)
Rank 2: Junior Data Analyst (500 XP)
Rank 3: Data Analyst (1,500 XP)
Rank 4: Senior Analyst (3,500 XP)
Rank 5: Lead Analyst (7,000 XP)

Rank 6: VP Structured Finance (15,000 XP)
Rank 7: Director of Analytics (25,000 XP)
Rank 8: Chief Risk Officer (40,000 XP)

B3. BUSINESS MODEL

Revenue Streams:

Stream 1: Banks and NBFCs (B2B Subscription)
Target: Banks, NBFCs, Rating Agencies
Price: ₹12,000 per analyst per year
Value: Train analysts before live portfolio work.
Avoid costly mistakes on real portfolios.
Target customers: Bajaj Finance, HDFC Bank,
CRISIL analysts, Small Finance Banks.

Stream 2: Individual Students (Retail)
Target: MBA Finance, CA, CFA students
Price: ₹4,999 one-time fee
Value: Job-ready skills + Zetheta Certification.
Platform access for 6 months.

Stream 3: University Licensing
Target: MBA Finance programs, CA Institutes
Price: ₹3 to 5 lakh per institution per year
Value: Practical curriculum supplement.
Faculty dashboard for student tracking.

Stream 4: Corporate Hiring Assessment
Target: Banks hiring analytics teams
Price: ₹75,000 per hiring campaign
Value: Objective skill assessment of candidates.
Replaces subjective technical interviews.

5-Year Revenue Projection:

Year 1: ₹16 Lakh
(5 banks × 10 users + 200 students)

Year 2: ₹1.8 Crore
(15 banks + 500 students + 3 universities)

Year 3: ₹5.2 Crore
(30 banks + 2,000 students + 10 universities)

Year 4: ₹12 Crore
(45 banks + 5,000 students + 20 universities)

Year 5: ₹22 Crore
(60+ institutions + 10,000 students + 35 universities)

Competitive Advantage:
→ Only platform combining DAX + Securitisation
→ India-specific (RBI rules, CRISIL, NACH, SMA)
→ Real simulated loan-level data
→ Time-pressured realistic scenarios
→ Formal Zetheta certification on completion

PART C: CASE STUDIES AND REAL-WORLD APPLICATIONS

C1. THE 2008 GLOBAL FINANCIAL CRISIS

Background:

Between 2003 and 2007, US banks gave mortgage loans to millions of subprime borrowers. These were people with poor credit scores, unverified incomes, and minimal down payments.

Banks immediately securitised these loans and sold them as AAA-rated securities to investors worldwide. Rating agencies rated them AAA because their models assumed US house prices would never fall nationwide.

The Analytics Failures:

Failure 1: No Vintage Analysis
Banks did not track loan performance by origination cohort. Loans given in 2006-2007 were of much lower quality than 2003-2004 loans. This deterioration was hidden in pool averages.

If Dashboard 3 (Vintage Analysis) had existed:
The 2006 vintage CNL at 12 months was estimated 3 to 4 times worse than the 2003 vintage. This would have been immediately visible on the cumulative loss curve chart.

Failure 2: No Roll Rate Monitoring
Nobody systematically tracked the transition
of loans from 30 DPD to 60 DPD to 90 DPD.
In early 2007, the roll rate from 60-89 DPD
to 90+ DPD jumped from 15% to 45% per month.

If Dashboard 2 (DPD Analysis) had existed:
The roll rate heatmap would have shown red cells
in the 60-89 DPD to 90+ DPD transition.
Early warning would have been generated
6 months before actual crisis.

Failure 3: Wrong Stress Testing
Models only stress-tested 5-10% house price
decline. Actual decline was 30-40%.

If Dashboard 6 (Stress Testing) had existed:
30% decline scenario would have shown
complete credit enhancement wipeout.
Structure would have been redesigned.

Failure 4: Geographic Concentration Ignored
60% of subprime loans were concentrated in
just 4 states: California, Florida, Nevada,
and Arizona. These were the most overheated
housing markets.

If Dashboard 1 (Executive Summary) had existed:
The HHI concentration index would have flagged
dangerous geographic concentration.
State distribution pie chart would have shown
4 states dominating the pool.

The Collapse:

House prices fell 30-40% across America.
Default rates reached 30% (models expected 3%).
AAA tranche investors began losing money.
Lehman Brothers collapsed on September 15, 2008.
Global credit markets completely froze.
The crisis spread to every country worldwide.

India's Response:
RBI introduced MRR and MHP requirements.

Originators must now retain 10% of each tranche.
Loans must be held 12 months before securitising.
This prevents repeat of US incentive problems.

C2. THE IL&FS CRISIS 2018

Background:

Infrastructure Leasing and Financial Services (IL&FS) was one of India's largest infrastructure finance companies with ₹91,000 crore in total debt across 347 group entities.

IL&FS was considered safe and was AAA rated.
In September 2018, it defaulted on its commercial paper obligations, shocking India's financial markets completely.

How Securitisation Was Affected:

IL&FS group entities had securitised infrastructure project receivables as Pass-Through Certificates (PTCs).

When underlying projects stopped generating revenue, the SPV cash flows dried up.
PTC investors stopped receiving monthly payments.
Mutual funds holding these PTCs faced massive investor redemption pressure.
Mutual funds sold healthy assets to meet redemptions, spreading the crisis to otherwise healthy NBFCs and bonds.

India's credit markets froze for several months.

The Analytics Failures:

Failure 1: Counterparty Concentration
Multiple securitisation pools had significant exposure to IL&FS group entities as borrowers.
Single group had 15-20% of some pool balances.

If Dashboard 1 had been used properly:
HHI concentration index would have flagged
dangerous single-name concentration.
"Top 10 borrower" table would have shown
IL&FS group dominating the pool.

Failure 2: Collection Efficiency Not Monitored
Monthly collection efficiency was declining
from 98% to 85% in the months before default.
This decline was not being tracked in real time.

If Dashboard 5 (Waterfall) had been used:
Collection efficiency declining below 95%
would have triggered an automated alert.
Recovery actions would have started months earlier.

Failure 3: DSCR Monitoring Absent
For infrastructure loans, DSCR (Debt Service
Coverage Ratio) is the key metric.
DSCR below 1.0 means project cannot pay its debt.
DSCR was declining from 1.5x to below 1.0x
in the quarters before default.

If this metric had been tracked:
Dashboard would have shown DSCR trend declining.
Early warning action would have been possible.

RBI Response:

The September 2021 revised securitisation framework
directly addressed IL&FS-type concentration risks:
→ Enhanced counterparty concentration limits
→ Stricter disclosure requirements for investors
→ Standardised monthly pool performance reporting
→ Trustee oversight requirements strengthened
→ Asset quality monitoring requirements added

PART D: PROJECT METHODOLOGY AND DELIVERABLES

D1. DAY-BY-DAY PROGRESS

Phase 1: Foundation Learning (Days 1-3)

Day 1:
Studied RBI 2021 Securitisation Framework.
Read rating agency methodologies for auto ABS.
Understood the 7-step securitisation process.
Learned about tranches, SPV, and waterfall.
Output: Securitisation landscape summary.

Day 2:
Opened all 4 Excel datasets.
Mapped every column to its business meaning.
Calculated basic pool statistics manually in Excel:
Total balance, loan count, DPD distribution.
Output: Manual calculation workbook.

Day 3:
Learned DPD bucket definitions and RBI SMA rules.
Understood ECL components: PD, LGD, EAD.
Learned waterfall mechanics: OC test, IC test.
Studied vintage analysis and static pool concept.
Output: Complete concept understanding.

Phase 2: Data Setup (Days 4-5)

Day 4:
Installed Power BI Desktop and DAX Studio.
Imported all 4 Excel files into Power BI.
Created GitHub repository (private).
Output: Power BI file with 4 tables loaded.

Day 5:
Created DimDate calendar table using DAX.
Built star schema with 5 relationships.
Verified all connections using test measures.
Wrote first 10 basic DAX measures.
Output: Complete data model with relationships.

Phase 3: DAX Development (Days 6-8)

Day 6:
Learned CALCULATE, SUMX, VAR, DIVIDE patterns.
Wrote all weighted average measures:
WAC, WAM, WALA, WA LTV, WA CIBIL.
Wrote all DPD rate measures for each bucket.
Output: 30+ working measures.

Day 7:
Learned time intelligence functions.
Wrote rolling averages and trend measures.
Wrote period-over-period comparison measures.
Wrote DPD trend direction indicator.
Output: Complete measure library.

Day 8:
Created DPD_Bucket calculated column.
Created IFRS9_Stage calculated column.
Wrote all IFRS 9 ECL measures.
Cross-validated ECL against Excel calculation.
Output: ECL measures validated.

Phase 4: Dashboard Development (Days 9-16)

Day 9-10: Dashboard 1 - Executive Summary
Built 8 KPI cards with conditional formatting.
Built DPD distribution bar chart with colors.
Built state distribution pie chart.
Built loan trend line chart.
Built top states table and OC gauge.

Day 11-12: Dashboard 2 - DPD Analysis
Built 4 DPD KPI cards with trend arrows.
Built stacked bar chart for monthly DPD.
Built 30+ DPD trend line with moving average.
Built roll rate heatmap matrix.
Built detailed 60+ DPD loan table.
Added all slicers and filters.

Day 13-14: Dashboard 3 - Vintage Analysis
Built cumulative loss curves for 15 vintages.
Built 12-month CNL comparison bar chart.

Built 24-month CNL comparison bar chart.
Built complete vintage summary table.
Added vintage slicer.

Day 15-16: Dashboard 4 - IFRS 9 ECL
Built Stage classification calculated column.
Built two donut charts for count and balance.
Built 7 ECL KPI cards with conditional colors.
Built ECL computation matrix table.
Built stacked bar for balance over time.

Day 17-18: Dashboard 5 - Cash Flow Waterfall
Built waterfall chart for cash flow steps.
Built trigger compliance status table.
Built cash flow allocation summary.
Built excess spread gauge.

Phase 5: Documentation (Days 19-20)

Day 19:
Wrote complete project report.
Created 10-slide presentation.
Built DAX measure dictionary.

Day 20:
Final review and quality check.
GitHub upload and repository transfer.
Submission preparation.

D2. CHALLENGES AND SOLUTIONS

Challenge 1: Understanding Securitisation
I had no prior knowledge of banking or finance.
Solution: Studied each concept before building it.
Read the RBI framework and used AI tools for
explanations. Built understanding from scratch.

Challenge 2: DAX Context Errors
Early measures gave wrong results due to
incorrect filter context handling.
Solution: Used VAR to debug step by step.
Tested each measure with a card visual.
Compared results with manual Excel calculations.

Challenge 3: Roll Rate Matrix
The heatmap matrix was not showing correct transition probabilities initially.
Solution: Verified that DPD_Bucket_Prior column existed in DPD Snapshot data with correct values. Adjusted the measure logic accordingly.

Challenge 4: ECL Numbers Not Matching
Stage 2 ECL was not matching Excel crosscheck.
Solution: Discovered PD must be lifetime PD (not annual PD) for Stage 2 provisioning. Corrected the PD assumption in the formula.

Challenge 5: Date Relationships
Time intelligence measures returned blank values when first tested.
Solution: Marked DimDate as Date Table in Table Tools. Verified all 5 relationships were correctly set as One-to-Many.

Challenge 6: DPD Sorting in Charts
DPD buckets were sorting alphabetically instead of in severity order on charts.
Solution: Added numbers (1-, 2-, 3-) at start of DPD bucket names to force correct sort order.

PART E: NO-CODE TOOLS AND PLATFORMS USED

E1. Microsoft Power BI Desktop
Purpose: Main dashboard development platform

How it was used:
→ Imported all 4 Excel datasets
→ Built star schema data model
→ Wrote 50+ DAX measures
→ Created 5 interactive dashboards
→ Applied conditional formatting
→ Built slicers for interactive filtering

Key features used:
→ Get Data: Excel import
→ Model View: Relationship building
→ DAX formula bar: Measure writing

→ Format pane: Visual formatting
→ Analytics pane: Reference lines
→ New Parameter: What-if sliders

E3. Microsoft Excel
Purpose: Validation and documentation

How it was used:
→ Manual ECL crosscheck calculation
→ Waterfall step validation
→ DAX measure dictionary documentation
→ Basic pool statistics for comparison

Essential Excel functions used:
SUMPRODUCT: Weighted average calculations
IF/Nested IF: Stage classification testing
COUNTIF/SUMIF: DPD bucket counting
VLOOKUP: PD and LGD parameter lookup

E4. GitHub
URL: Udhaya1075 (Udhai) · GitHub (Free)
Purpose: Version control and submission

How it was used:
→ Private repository for all project files
→ Daily commits throughout project
→ README.md documentation
→ Final repository transfer to ZethetaIntern

E5. Google Docs
URL:
https://docs.google.com/document/d/18iPN7u_geeT8YKTSdIfVyrxZXZUI7aBmNlBs-rdxHhk/edit?usp=sharing (Free)
Purpose: Report writing and documentation

How it was used:
→ Drafted all report sections progressively
→ Automatic table of contents generation
→ Screenshot insertion for dashboards
→ Exported as PDF for submission

E6. Claude.ai (AI Research Tool)
Purpose: Learning support and concept clarification

How it was used:
→ Understanding securitisation concepts
→ Initial DAX formula structuring and ideas
→ Explaining RBI regulatory requirements
→ All AI outputs were verified in Power BI
before being used in final submission

Important note: Claude.ai was used as a learning accelerator and research tool, not as a replacement for understanding. Every DAX measure was tested and verified independently in Power BI.

PART F: SUBMISSION ARTIFACTS

Complete list of files submitted:

1. Power BI Dashboard File
   Filename: 201169C_Udhai_ZeThatA.pbix
   Contents: 5 interactive dashboards,
   50+ DAX measures, complete data model
   with star schema and 5 relationships

2. Main Project Report
   Filename: 201169C_Udhai_Securitisation AI Agent - Final Report.pdf
   Contents: All 6 parts, dashboard screenshots,
   case studies, methodology, 40+ pages

3. DAX Measure Dictionary
   Filename: 201169C_Udhai_DAX_Dictionary.xlsx
   Contents: All 50+ measures documented with
   formula, business purpose, and category

4. GitHub Repository
   URL: github.com/ZethetaIntern/[your-repo]
   Contents: All files above plus README.md
   with project overview and instructions

APPENDIX A: DASHBOARD SCREENSHOTS

Figure 1: Dashboard 1 - Executive Summary

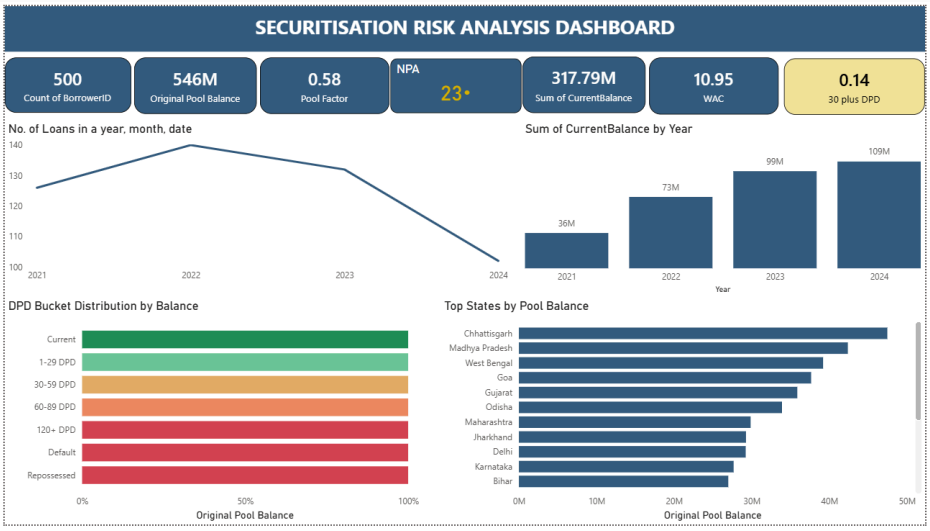


Figure 2: Dashboard 2 - DPD Analysis

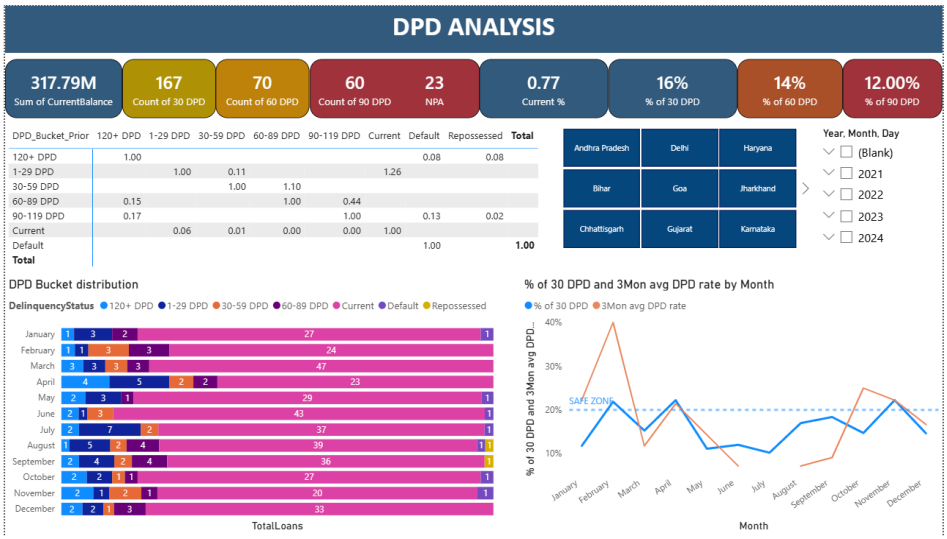


Figure 3: Dashboard 3 - Vintage Analysis

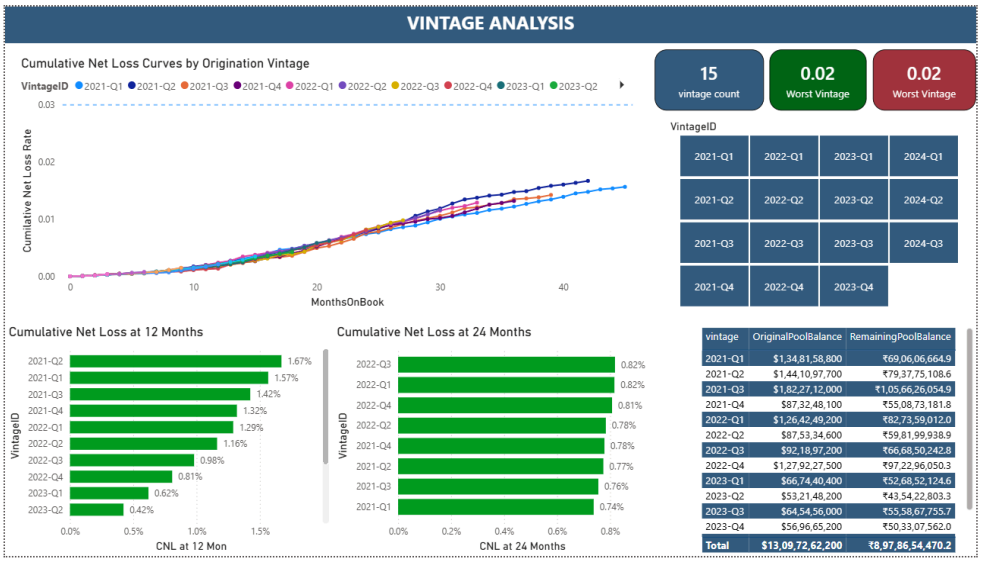


Figure 4: Dashboard 4 - IFRS 9 ECL

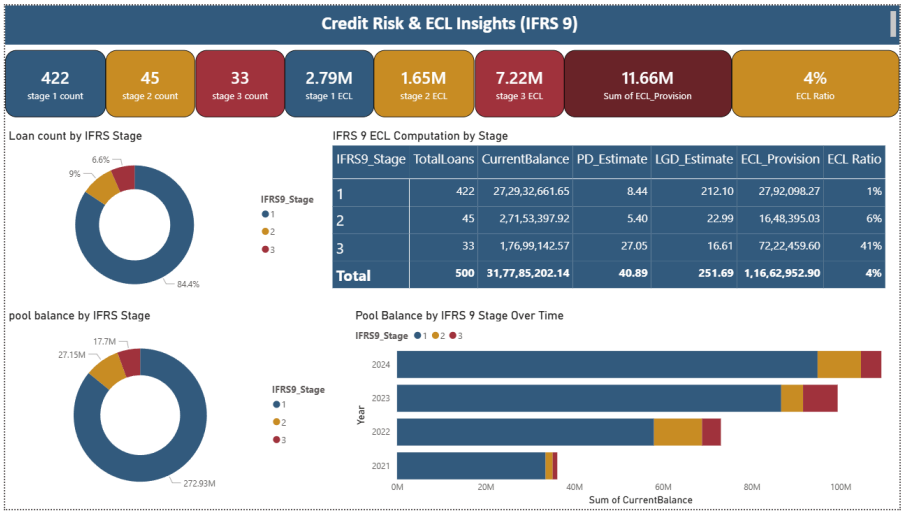


Figure 5: Dashboard 5 - Cash Flow Waterfall

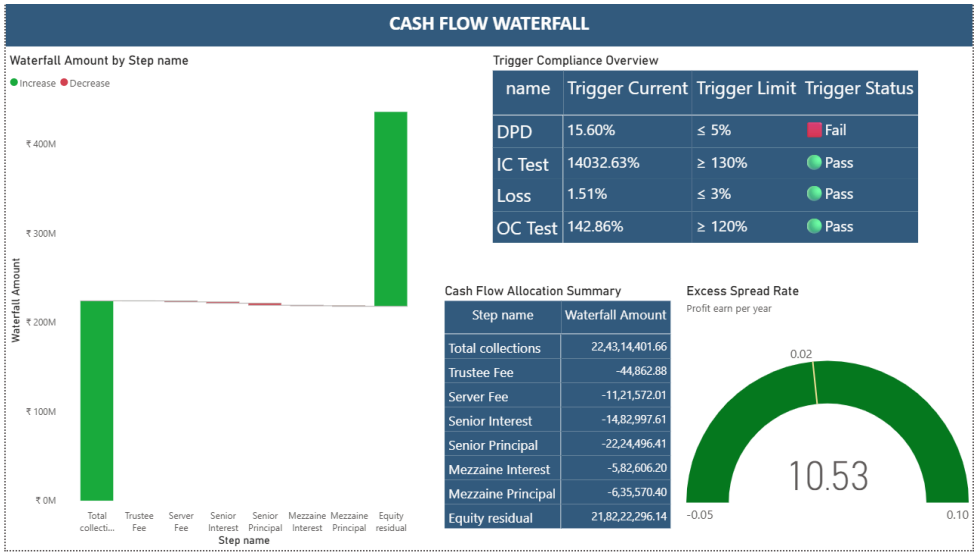


Figure 6: Power BI Data Model

